

Full Length Article

On the approximation capability of shallow and deep neural networks having smooth activations with respect to the Sobolev norm

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ARTICLE INFO

Keywords:

Deep learning
Neural networks
Smooth activation functions
Universal approximation
Sobolev norm

ABSTRACT

In this paper, we investigate the simultaneous approximation of the functions and their derivatives by neural networks having smooth non-polynomial activation functions and a fixed depth, which is motivated by the physics-informed machine learning. We start by proving that the neural networks with smooth non-polynomial activation functions and with only one hidden layer having width $\mathcal{O}(N^d)$ can approximate any $W^{s,p}$ -regular function with rate $\mathcal{O}(N^{k-s})$ in the $W^{k,p}$ -norm. We then extend this result to the networks having more than one hidden layers by using the mathematical induction.

1. Introduction

In recent years, the deep neural networks has been received a huge attention due to their remarkable performance in various areas of engineering and science, including computer vision, natural language processing, recommendation systems, robotics, scientific computing, and so on (see (Cuomo et al., 2022; Goodfellow et al., 2016; LeCun et al., 2015) and the references therein). Their success in practice has demonstrated that the deep learning can be a powerful tool for solving complex or high-dimensional problems which were previously considered almost intractable in many traditional methods.

Nevertheless, there is still a lack of rigorous analysis of the reason why they perform so well. Among the theoretical studies to address this, the universal approximation capability of deep neural networks is regarded as one of the most important and fundamental topics, which asserts that the neural networks can approximate a broad class of functions in arbitrary accuracy under suitable conditions.

The research on the approximation capability of neural networks can be traced back to the 1980s (Cybenko, 1989; Hornik et al., 1989), in which it was proved that any continuous function can be approximated by the shallow neural networks (i.e., the neural networks having only one hidden layers) with sigmoidal activation functions with respect to the L^p -norm for $1 \leq p \leq \infty$ over an arbitrary compact subset of a Euclidean space. Later, Mhaskar (1996) proved the optimal approximation capability of shallow neural networks for the functions in the Sobolev space with respect to the L^p -norm, i.e., given a $W^{s,p}$ -regular function over a compact subset of $\mathbb{R}^d$ with integer $s > 0$, for any positive integer N there exists a shallow neural network having width $\mathcal{O}(N^d)$ such that the error in the L^p -norm over the compact set can be bounded above by $\mathcal{O}(N^{-s})$. More recently, many authors have considered more general

settings, such as different activations including ReLU and its variants (Leshno et al., 1993; Mao & Zhou, 2023; Siegel & Xu, 2020; Yarotsky, 2017), fixed depths (De Ryck et al., 2021; DeVore et al., 2021; Guliyev & Ismailov, 2018), deep and narrow networks (Kidger & Lyons, 2020; Park et al., 2021), and approximation on unbounded domains (Abdeljawad & Dittrich, 2024; Wang & Qu, 2022).

However, there are relatively fewer studies on the simultaneous approximation by neural networks to not only the functions but also their derivatives. Some of the exceptions can be found in (Belomestny et al., 2023; De Ryck et al., 2021; Hon & Yang, 2022; Hornik et al., 1990; Li, 1996; Xu & Cao, 2005) and the references therein. These results are important for the models involving the derivatives of neural networks, such as the physics-informed machine learning (E & Yu, 2018; Raissi et al., 2019), Sobolev training (Czarnecki et al., 2017; Son et al., 2023) and gradient-penalized generative adversarial networks (Gulrajani et al., 2017; Mroueh et al., 2018).

In this paper, we investigate the approximation capability of the neural networks with respect to the Sobolev norm. Specifically, we focus on the networks having smooth activation functions, fixed depth and large width, which is widely used in the physics-informed neural networks (De Ryck et al., 2021; Lu et al., 2018; Raissi et al., 2019). Our main contributions are summarized as follows: Given any function in $W^{s,p}([-1, 1]^d)$ with $s \geq 0$ and $1 \leq p \leq \infty$,

- (Theorem 1) for each $N \in \mathbb{N}$ there exists a shallow neural network approximation, having width $\mathcal{O}(N^d)$ and smooth non-polynomial activation function, such that the $W^{k,p}$ -norm error is of rate $\mathcal{O}(N^{k-s})$ for any integer $k \in [0, s]$;
- (Theorem 2) for any $N \in \mathbb{N}$ and any $L \geq 2$, there exists a deep neural network approximation with non-polynomial smooth activation

E-mail address: hjpark2357@korea.ac.kr<https://doi.org/10.1016/j.neunet.2026.108935>

Received 25 July 2025; Received in revised form 14 February 2026; Accepted 2 April 2026

Available online 3 April 2026

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Table 1

Comparison of some results on the simultaneous approximation by deep neural networks.

Paper	Function Class	Norm	Activation	Depth	Width	Error Bound	Remark
Siegel and Xu (2020)	$\mathcal{B}^{k+1}$	$W^{k,2}$	$W^{k,\infty}$ polynomial decay	2	$\mathcal{O}(N)$	$\mathcal{O}(N^{-1/2})$	Barron space
De Ryck et al. (2021)	$W^{s,p}$	$W^{k,p}$	tanh	3	one $\mathcal{O}(N^d)$ the other $\mathcal{O}(N)$	$\mathcal{O}((\log N)^k N^{k-s})$	
Gühring and Raslan (2021)	$W^{s,p}$	$W^{k,p}$	PU-decay	$\mathcal{O}(1)$	$\mathcal{O}(N^d)$	$\mathcal{O}(N^{k-s})$	encodable weights
Hon and Yang (2022)	C^s	$W^{k,p}$	ReLU or ReQU	$\mathcal{O}(L^d \log L)$	$\mathcal{O}(N^d \log N)$	$\mathcal{O}((NL)^{-2(k-s)})$	$1 \leq p < \infty$
Belomestny et al. (2023)	$\mathcal{H}^s$	$\mathcal{H}^k$	ReQU	$\mathcal{O}(1)$	$\mathcal{O}(N^d)$	$\mathcal{O}(N^{k-s})$	Hölder space
This	$W^{s,p}$	$W^{k,p}$	smooth non-polynomial	≥ 2	one $\mathcal{O}(N^d)$ the others 2	$\mathcal{O}(N^{k-s})$	

function and with L hidden layers, one layer having width $\mathcal{O}(N^d)$ and the others having width $\mathcal{O}(1)$, such that the $W^{k,p}$ -norm error is of rate $\mathcal{O}(N^{k-s})$ for any integer $k \in [0, s]$.

A key idea to prove Theorem 1 is to employ the de la Vallée Poussin-polynomial approximation and construct a shallow neural network approximation of each monomial term in the polynomial by using the central finite difference formula. This idea was also used in (Leshno et al., 1993; Mhaskar, 1996), but the authors only considered the L^p -norm error bounds. For Theorem 2, we use the mathematical induction on L . Namely, given a neural network approximation having ℓ hidden layers, one can construct a network approximation having $\ell + 1$ hidden layers by using the central finite difference formula as in Theorem 1.

Let us briefly compare our results with other related works (see Table 1). In (De Ryck et al., 2021), the authors constructed a tanh neural network approximation having width at most $\mathcal{O}(N^d)$, but the number of hidden layers is fixed at 2 and the derived $W^{k,p}$ -norm error bound is $\mathcal{O}((\log N)^k N^{k-s})$. Gühring and Raslan (2021) constructed a neural network approximation having encodable weights and a similar error bound with ours, but the depth was not explicitly given. Siegel and Xu (2020) proved a dimensionless approximation by shallow neural networks with polynomial decay activation having width $\mathcal{O}(N)$ and error bound $\mathcal{O}(N^{-1/2})$, but the target function needs to be contained in the Barron space, which is smaller than the Sobolev space of the same order. One can also find the results on the simultaneous approximation by ReLU and ReQU neural networks, including (Belomestny et al., 2023; Hon & Yang, 2022). Belomestny et al. (2023) constructed ReQU neural network approximations having width $\mathcal{O}(N^d)$, depth $\mathcal{O}(1)$ and error bound $\mathcal{O}(N^{k-s})$, while Hon and Yang (2022) constructed ReLU and ReQU neural network approximations having width $\mathcal{O}(N \log N)$, depth $\mathcal{O}(L \log L)$ and better error bound $\mathcal{O}((NL)^{-2(k-s)/d})$. However, the target function space in their results is the C^s -space or the Hölder space, which are smaller than the Sobolev space of the same order.

On the other hand, the main features of our results are as follows: (i) The neural network approximation can have arbitrary depth, including the case of only one hidden layer. (ii) Any smooth non-polynomial real-valued function on $\mathbb{R}$ can be used as an activation function. (iii) The error bound is independent of the activation. (iv) The $W^{k,p}$ -norm error upper bound $\mathcal{O}(N^{k-s})$ is almost optimal in the sense of (Gühring & Raslan, 2021). (v) The target function space is the Sobolev space, which usually appears in the theory of PDEs, and so our results are expected to be applicable for the theoretical analysis for the physics-informed machine learning. Therefore, we believe that our results provide an improvement over the available results for the approximation capability of smooth neural networks having relatively small depth.

The rest of the paper is organized as follows. In Section 2, several preliminary definitions and notations are provided. In Section 3, we state our main theorems. We then prove the theorems in Section 4. The conclusion follows in Section 5.

2. Preliminaries

We summarize some basic notations and definitions. Let $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$ and $d \in \mathbb{N}$. Given a multi-index $\mathbf{n} = (n_1, \dots, n_d) \in \mathbb{N}_0^d$ and a real vector

$$\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d,$$

$$|\mathbf{n}| := |n_1| + \dots + |n_d|, \quad \mathbf{x}^{\mathbf{n}} = x_1^{n_1} \dots x_d^{n_d}, \quad \partial_{\mathbf{x}}^{\mathbf{n}} = \frac{\partial^{|\mathbf{n}|}}{\partial x_1^{n_1} \dots \partial x_d^{n_d}}.$$

Given multi-indices $\mathbf{n} = (n_1, \dots, n_d)$ and $\mathbf{m} = (m_1, \dots, m_d)$, let

$$\binom{\mathbf{n}}{\mathbf{m}} := \binom{n_1}{m_1} \dots \binom{n_d}{m_d} = \prod_{i=1}^d \frac{n_i!}{m_i!(n_i - m_i)!}.$$

In addition, we write $\mathbf{n} \leq \mathbf{m}$ if $n_i \leq m_i$ for any $i = 1, \dots, d$. Similarly, for $n \in \mathbb{N}_0$ we write $\mathbf{n} \leq n$ (resp. $\mathbf{n} \geq n$) if $n_i \leq n$ (resp. $n_i \geq n$) for any $i = 1, \dots, d$.

We follow the usual notation and definition of Sobolev spaces (see, e.g., (Evans, 2010; Grisvard, 2011; Wloka, 1987)). Given an open set $O \subseteq \mathbb{R}^d$ with $d \in \mathbb{N}$, $1 \leq p \leq \infty$ and $s \geq 0$, let $\|\cdot\|_{s,p,O}$ (resp. $|\cdot|_{s,p,O}$) denote the norm (resp. seminorm) of the Sobolev space $W^{s,p}(O)$. Conventionally, we let $W^{0,p}(O) = L^p(O)$. Given $k \in \mathbb{N}$, $u \in W^{k,p}(O)$ and a multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq k$, let

$$\partial^{\mathbf{k}} u := \frac{\partial^{|\mathbf{k}|} u}{\partial x_1^{k_1} \dots \partial x_d^{k_d}}.$$

We also introduce the periodic function spaces on $\mathbb{T}^d$ with $\mathbb{T} := [-\pi, \pi]$. For $1 \leq p \leq \infty$ and $s \geq 0$, let $W_*^{s,p}(\mathbb{T}^d)$ be the subspace of all coordinate-wise 2π -periodic functions in $W^{s,p}(\mathbb{T}^d)$, and let $L_*^p(\mathbb{T}^d) = W_*^{0,p}(\mathbb{T}^d)$. Also, given $n \in \mathbb{N}_0 \cup \{\infty\}$, let $C_*^n(\mathbb{T}^d)$ denote the subspace of all coordinate-wise 2π -periodic functions in $C^n(\mathbb{T}^d)$.

Given $u \in L_*^p(\mathbb{T}^d)$, its Fourier coefficients are

$$\hat{u}(\mathbf{k}) := (2\pi)^{-d} \int_{\mathbb{T}^d} u(\mathbf{z}) e^{-i\mathbf{k} \cdot \mathbf{z}} d\mathbf{z}, \quad \mathbf{k} \in \mathbb{Z}^d,$$

and given $\mathbf{n} \in \mathbb{N}_0^d$, the partial sum of the Fourier series of u is defined by

$$(S_{\mathbf{n}} u)(\mathbf{x}) := \sum_{0 \leq \mathbf{k} \leq \mathbf{n}} (\hat{u}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}} + \hat{u}(-\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{x}}), \quad \mathbf{x} \in \mathbb{T}^d.$$

We finally introduce the notion of deep neural networks. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ denote an activation function. Given $L \in \mathbb{N}$ and $d \in \mathbb{N}$, a neural network with L hidden layers is a function $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$\Phi(\mathbf{x}) = (A_L \circ \sigma \circ \dots \circ A_1 \circ \sigma \circ A_0)(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^d,$$

where, for each $\ell = 0, 1, \dots, L$, the function A_ℓ is the affine mapping from $\mathbb{R}^{n_\ell}$ into $\mathbb{R}^{n_{\ell+1}}$ (with $n_{L+1} = 1$ and $n_0 = d$), i.e.,

$$A_\ell(\mathbf{z}) = \overline{\mathbf{W}}_\ell \mathbf{z} + \mathbf{b}_\ell, \quad \mathbf{z} \in \mathbb{R}^{n_{\ell-1}},$$

with $\overline{\mathbf{W}}_\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$ and $\mathbf{b}_\ell \in \mathbb{R}^{n_{\ell+1}}$, and where the activation function σ is applied coordinatewise. For each $\ell = 0, 1, \dots, L$, n_ℓ is called the width of ℓ -th layer.

3. Main results

In this section, we state our main results. The proof will be given in Section 4.

Theorem 1. Let σ be a non-polynomial, smooth and real-valued function on $\mathbb{R}$. Let $f \in W^{s,p}(\Omega)$ with $\Omega = [-1, 1]^d$, $d \in \mathbb{N}$, $s \geq 0$ and $1 \leq p \leq \infty$. Then, for each $N \in \mathbb{N}$, there exists a neural network f_N with activation σ and with one hidden layer having width at most $(4N + 1)^d$ such that

$$\|f - f_N\|_{k,p,\Omega} \leq C(s, p, d) N^{k-s} \|f\|_{s,p,\Omega}, \quad \forall k \in \mathbb{N}_0 \text{ with } k \leq s, \quad (1)$$

where $C(s, p, d)$ is a positive constant depending only on s, p and d . Furthermore, for any $\epsilon > 0$ there exists a neural network f_ϵ with activation σ and with one hidden layer such that

$$\|f - f_\epsilon\|_{s,p,\Omega} \leq \epsilon. \quad (2)$$

Theorem 2. Let σ be a non-polynomial, smooth and real-valued function on $\mathbb{R}$. Let $f \in W^{s,p}(\Omega)$ with $\Omega = [-1, 1]^d$, $d \in \mathbb{N}$, $s \geq 0$ and $1 \leq p \leq \infty$. Then, for each $N \in \mathbb{N}$, there exists a neural network f_N with activation σ and with $L \geq 2$ hidden layers, one having width at most $(4N + 1)^d$ and the others having width at most 2, such that

$$\|f - f_N\|_{k,p,\Omega} \leq C(s, p, d) N^{k-s} \|f\|_{s,p,\Omega}, \quad \forall k \in \mathbb{N}_0 \text{ with } k \leq s, \quad (3)$$

where $C(s, p, d)$ is a positive constant depending only on s, p and d . Furthermore, if $s \in \mathbb{N}_0$, then for any $\epsilon > 0$ there exists a neural network f_ϵ with activation σ and with $L \geq 2$ hidden layers such that

$$\|f - f_\epsilon\|_{s,p,\Omega} \leq \epsilon. \quad (4)$$

It is also possible to construct a neural network approximation with exponential rates in the Sobolev norm for a class of analytic functions.

Theorem 3. Let σ be a non-polynomial, smooth and real-valued function on $\mathbb{R}$. Let f be analytic on $\Omega = [-1, 1]^d$ with $d \in \mathbb{N}$ such that there exists a constant $R_f \geq 1$ satisfying $|f|_{n,\infty,\Omega} \leq R_f^n$ for any $n \in \mathbb{N}_0$. Also, let $s \in \mathbb{N}_0$. Then, for any $N \in \mathbb{N}$, there exists a neural network f_N with smooth non-polynomial activation σ and with $L \geq 1$ hidden layers, one having width at most $(2N + 2s + 6\lceil dR_f \rceil + 1)^d$ and the others having width at most 2, such that

$$\|f - f_N\|_{s,\infty,\Omega} \leq C(s, d, R_f) \exp(-N),$$

where $C(s, d, R_f)$ is a positive constant depending only on s, d and R_f .

Remark 1. Although we have assumed that $\Omega = [-1, 1]^d$, our results can be easily extended to the case when Ω is a bounded Lipschitz domain in $\mathbb{R}^d$.

Remark 2. The following popular activation functions can be used in our results:

- sigmoid (logistic): $\sigma(x) = (1 + \exp(-x))^{-1}$;
- hyperbolic tangent: $\sigma(x) = \tanh(x)$;
- arctangent: $\sigma(x) = \arctan(x)$;
- Gaussian: $\sigma(x) = \exp(-x^2)$;
- Softplus (Glorot et al., 2011): $\sigma(x) = \log(1 + \exp(x))$;
- SiLU (Hendrycks & Gimpel, 2016): $\sigma(x) = x(1 + \exp(-x))^{-1}$.

Remark 3. Given $s \geq 0$, the Barron space $\mathcal{B}^s$ is defined by

$$\mathcal{B}^s = \{v \in \mathcal{S}(\mathbb{R}^d) : \|v\|_{\mathcal{B}^s} < \infty\}, \quad \|v\|_{\mathcal{B}^s} = \int_{\mathbb{R}^d} (1 + |\xi|)^s |\hat{v}(\xi)| d\xi,$$

where $\mathcal{S}(\mathbb{R}^d)$ is the space of tempered distributions on $\mathbb{R}^d$ and $\hat{v}$ denotes the Fourier transform of v . Siegel and Xu (2020) established approximation results by shallow neural networks for the Barron space $\mathcal{B}^{m+1}$ with an integer $m \geq 0$. Specifically, given $f \in \mathcal{B}^{m+1}$ and an activation function $\sigma \in W^{m,\infty}(\mathbb{R})$ satisfying a polynomial decay condition, for any $n \in \mathbb{N}$ there exists a shallow neural network f_n of width n such that

$$\|f - f_n\|_{m,2,\Omega} \leq C(\sigma) n^{-\frac{1}{2}} \|f\|_{\mathcal{B}^{m+1}}, \quad (5)$$

where $C(\sigma)$ is a positive constant depending only on σ . Since $\mathcal{B}^{m+1}$ is continuously embedded in $W^{m+1,2}(\Omega)$, Theorem 1 covers a broader function space of the same order than the one given in (5), but the width of the neural networks scales $\mathcal{O}(n^d)$.

Nevertheless, we can derive a similar result for a subspace of the Barron space $\mathcal{B}^s$ with $s \geq 0$. Note that, according to Theorem 4.3 in (Meng & Ming, 2022), for any $s \geq 0$ and $\alpha > s + d/2$,

$$W^{\alpha,2}(\mathbb{R}^d) \hookrightarrow \mathcal{B}^s$$

and the inclusion essentially fails for $\alpha = s + d/2$. Now let $s \geq 0$, $\alpha > s + d/2$, and let σ be a non-polynomial, smooth and real-valued function

on $\mathbb{R}$. Then, by Theorem 2, given $f \in W^{\alpha,2}(\mathbb{R}^d)$, there exists a sequence $\{f_n\}_n$ of neural networks with activation σ and with a finite number of hidden layers, one having width at most $\max\{n, 5^d\}$ and the others having width at most 2, such that, for any $k \in \mathbb{N}_0$ with $k \leq \alpha$,

$$\|f - f_n\|_{k,2,\Omega} \leq C_1 \left[\frac{n^{1/d} - 1}{4} \right]^{k-\alpha} \|f\|_{\alpha,2,\Omega} \leq C_2 n^{\frac{k-\alpha}{d}} \|f\|_{\alpha,2,\Omega},$$

where $C_1, C_2 > 0$ are constants depending only on α and d . Since $\alpha > s + d/2$, the inequality above implies that, for any integer $k \leq \alpha$,

$$\|f - f_n\|_{k,2,\Omega} \leq C n^{-\frac{1}{2}} \|f\|_{\alpha,2,\Omega},$$

where $C > 0$ is a constant depending only on α and d . Consequently, we obtain the convergence rate $\mathcal{O}(n^{-1/2})$ with respect to the width n , which is consistent with the one given in (5).

4. Proof of the main results

For simplicity of presentation, we write $X \lesssim Y$ or $Y \gtrsim X$ if there exists a positive constant C depending only on s, p and d . We also write $X \approx Y$ if both $X \lesssim Y$ and $Y \lesssim X$ hold.

4.1. Some technical lemmas

We first present preliminary lemmas to prove our main results. In the following lemma, we consider the trigonometric polynomial approximation of the functions in $W_*^{s,p}(\mathbb{T}^d)$ for $s \in \mathbb{N}$ and $1 \leq p \leq \infty$.

Lemma 1. For each $N \in \mathbb{N}$, let $\mathcal{V}_N$ be the N -th de la Vallée Poussin operator from $L_*^p(\mathbb{T}^d)$ into $C_*^\infty(\mathbb{T}^d)$ given by

$$\mathcal{V}_N u := (N+1)^{-d} \sum_{N \leq \mathbf{n} \leq 2N} S_{\mathbf{n}} u, \quad u \in L_*^p(\mathbb{T}^d).$$

Then we have, for any nonnegative integers k and s with $k \leq s$,

$$\|u - \mathcal{V}_N u\|_{k,p,\mathbb{T}^d} \lesssim N^{k-s} \|u\|_{s,p,\mathbb{T}^d}, \quad \forall u \in W_*^{s,p}(\mathbb{T}^d). \quad (6)$$

Proof. We first note that the following estimate holds (see, e.g., (Mhaskar, 1996; Timan, 1963)): For any integer $t \geq 0$,

$$\|v - \mathcal{V}_N v\|_{0,p,\mathbb{T}^d} \lesssim N^{-t} \|v\|_{t,p,\mathbb{T}^d}, \quad \forall v \in W_*^{t,p}(\mathbb{T}^d). \quad (7)$$

Let $u \in W_*^{s,p}(\mathbb{T}^d)$. Then, for any $\mathbf{x} \in \mathbb{T}^d$, $\mathbf{m} \in \mathbb{Z}^d$ and multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$,

$$\partial_{\mathbf{x}}^{\mathbf{k}} (\hat{u}(\mathbf{m}) e^{i\mathbf{m} \cdot \mathbf{x}}) = (i\mathbf{m})^{\mathbf{k}} \hat{u}(\mathbf{m}) e^{i\mathbf{m} \cdot \mathbf{x}} = \widehat{\partial^{\mathbf{k}} u}(\mathbf{m}) e^{i\mathbf{m} \cdot \mathbf{x}}.$$

This implies that, for any multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$,

$$\begin{aligned} \partial^{\mathbf{k}} \mathcal{V}_N u &= (N+1)^{-d} \sum_{N \leq \mathbf{n} \leq 2N} \partial^{\mathbf{k}} S_{\mathbf{n}} u \\ &= (N+1)^{-d} \sum_{N \leq \mathbf{n} \leq 2N} S_{\mathbf{n}} (\partial^{\mathbf{k}} u) = \mathcal{V}_N (\partial^{\mathbf{k}} u). \end{aligned}$$

Thus the operator $\mathcal{V}_N$ maps $\partial^{\mathbf{k}} u$ to $\partial^{\mathbf{k}} \mathcal{V}_N u$. Since $\partial^{\mathbf{k}} u \in W_*^{s-|\mathbf{k}|,p}(\mathbb{T}^d)$, we obtain by applying (7) with $v = \partial^{\mathbf{k}} u$ and $t = s - |\mathbf{k}|$ that

$$\begin{aligned} \|\partial^{\mathbf{k}} u - \partial^{\mathbf{k}} \mathcal{V}_N u\|_{0,p,\mathbb{T}^d} &= \|\partial^{\mathbf{k}} u - \mathcal{V}_N (\partial^{\mathbf{k}} u)\|_{0,p,\mathbb{T}^d} \\ &\lesssim N^{|\mathbf{k}|-s} \|\partial^{\mathbf{k}} u\|_{s-|\mathbf{k}|,p,\mathbb{T}^d} \lesssim N^{|\mathbf{k}|-s} \|u\|_{s,p,\mathbb{T}^d} \end{aligned}$$

for any multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$. Therefore (6) follows. $\square$

We next consider the approximation of the functions in $W^{s,p}(\Omega)$ with $s \geq 0$ by polynomials with respect to the norm $\|\cdot\|_{k,p,\Omega}$ with $0 \leq k \leq s$.

Lemma 2. Let $f \in W^{s,p}(\Omega)$ with $\Omega = [-1, 1]^d$, $d \in \mathbb{N}$, $s \geq 0$ and $1 \leq p \leq \infty$. Then, for each $N \in \mathbb{N}$, there exists a polynomial $P_N f$ on Ω with degree at most $2N$ in each variable such that

$$\|f - P_N f\|_{k,p,\Omega} \lesssim N^{k-s} \|f\|_{s,p,\Omega}, \quad 0 \leq k \leq s. \quad (8)$$

Moreover, for each multi-index $\mathbf{n}$ with $\mathbf{n} \leq 2N$, the $\mathbf{n}$ -th coefficient $a_{\mathbf{n}}$ of the polynomial $P_N f$ satisfies $|a_{\mathbf{n}}| \lesssim \|f\|_{0,p,\Omega}$.

Proof. We first consider the case when s is a nonnegative integer. In this case, although the proof is essentially the same with the proof of (3.14) in (Mhaskar, 1996), we provide it for the completeness. Let ψ be a smooth cut-off function such that $\psi = 1$ on Ω and $\psi = 0$ outside of $(-3/2, 3/2)^d$. By the Sobolev extension theorem, there exists an extension $\tilde{f} \in W^{s,p}(\mathbb{R}^d)$ of f such that $\|\tilde{f}\|_{s,p,\mathbb{R}^d} \lesssim \|f\|_{s,p,\Omega}$. Then $\tilde{f} := \psi \tilde{f}$ coincides with f on Ω , vanishes outside of $(-3/2, 3/2)^d$, and $\|\tilde{f}\|_{s,p,\mathbb{R}^d} \lesssim \|f\|_{s,p,\Omega}$. Now we define f^* by the coordinatewise 2π -periodic function such that

$$f^*(t_1, \dots, t_d) := \tilde{f}(2 \cos t_1, \dots, 2 \cos t_d), \quad \forall (t_1, \dots, t_d) \in \mathbb{R}^d. \quad (9)$$

Since the mapping

$$\mathfrak{Z} : \left[\frac{\pi}{3}, \frac{2\pi}{3} \right]^d \rightarrow [-1, 1]^d, \quad \mathfrak{Z}(t_1, \dots, t_d) = (2 \cos t_1, \dots, 2 \cos t_d),$$

is a smooth diffeomorphism such that $|\det D\mathfrak{Z}| \approx 1$ on $[\pi/3, 2\pi/3]^d$, where $D\mathfrak{Z}$ denotes the Jacobian matrix of $\mathfrak{Z}$, Theorem 3.41 in (Adams & Fournier, 2003) implies that

$$\|f^*\|_{\ell,p,\mathbb{T}^d} \approx \|f\|_{\ell,p,\Omega}, \quad \forall \ell = 0, \dots, s. \quad (10)$$

Let us first construct a trigonometric polynomial approximation $\mathcal{V}_N f^*$ of f^* . Note that, by the construction of f^* , it holds that

$$f^*(t_1, \dots, -t_i, \dots, t_d) = f^*(t_1, \dots, t_i, \dots, t_d), \quad \forall i = 1, \dots, d, \quad \forall \mathbf{t} = (t_1, \dots, t_d) \in \mathbb{R}^d.$$

Then the partial sum $S_{\mathbf{n}} f^*$ for $\mathbf{n} \in \mathbb{N}_0^d$ can be rewritten as

$$(S_{\mathbf{n}} f^*)(\mathbf{t}) = \sum_{0 \leq \mathbf{k} \leq \mathbf{n}} c_{\mathbf{k}}(f^*) \cos(k_1 t_1) \cdots \cos(k_d t_d), \quad \forall \mathbf{t} = (t_1, \dots, t_d) \in \mathbb{R}^d,$$

where $c_{\mathbf{k}}(f^*)$ is given by

$$c_{\mathbf{k}}(f^*) := \frac{2}{(2\pi)^d} \int_{\mathbb{T}^d} f^*(\mathbf{z}) \cos(k_1 z_1) \cdots \cos(k_d z_d) d\mathbf{z}.$$

For each $k \in \mathbb{N}_0$, let T_k be the k -th Chebyshev polynomial of the first kind, i.e., it is the polynomial of degree at most k such that

$$T_k(2 \cos t) = \cos(kt), \quad \forall t \in \mathbb{T}. \quad (11)$$

Given a multi-index $\mathbf{k} = (k_1, \dots, k_d) \in \mathbb{N}_0^d$, define $T_{\mathbf{k}}$ by the function on $\mathbb{T}^d$ such that

$$T_{\mathbf{k}}(x_1, \dots, x_d) = T_{k_1}(x_1) \cdots T_{k_d}(x_d), \quad \forall (x_1, \dots, x_d) \in \mathbb{R}^d.$$

Then we have, for any $\mathbf{t} = (t_1, \dots, t_d) \in \mathbb{T}^d$,

$$(\mathcal{V}_N f^*)(\mathbf{t}) = (N+1)^{-d} \sum_{N \leq \mathbf{n} \leq 2N} \sum_{0 \leq \mathbf{k} \leq \mathbf{n}} c_{\mathbf{k}}(f^*) T_{\mathbf{k}}(2 \cos t_1, \dots, 2 \cos t_d).$$

Let $P_N f$ be the polynomial of degree at most $2N$ in each variable defined by

$$(P_N f)(\mathbf{x}) = (N+1)^{-d} \sum_{N \leq \mathbf{n} \leq 2N} \sum_{0 \leq \mathbf{k} \leq \mathbf{n}} c_{\mathbf{k}}(f^*) T_{\mathbf{k}}(\mathbf{x}). \quad (12)$$

By (9) and (11), for any $\mathbf{t} = (t_1, \dots, t_d) \in \mathbb{T}^d$,

$$(\tilde{f} - P_N f)(2 \cos t_1, \dots, 2 \cos t_d) = (f^* - \mathcal{V}_N f^*)(t_1, \dots, t_d).$$

Then, as in (10), we obtain

$$\|f - P_N f\|_{k,p,\Omega} \approx \|f^* - \mathcal{V}_N f^*\|_{k,p,\mathbb{T}^d}, \quad \forall k \in \mathbb{N}_0 \text{ with } k \leq s.$$

Now combining this with (10) and Lemma 1, we obtain (8) when s is a nonnegative integer.

Now we consider the case when s is not an integer and $s > 0$. Let s_0 be the largest integer such that $s_0 < s$, and let $s_1 = s_0 + 1$. Also, let $\theta \in (0, 1)$ satisfy $s = (1 - \theta)s_0 + \theta s_1$. Note that the operator $T : f \mapsto f - P_N f$ satisfies

$$\|Tf\|_{k,p,\Omega} \lesssim N^{k-s_0} \|f\|_{s_0,p,\Omega}, \quad \forall f \in W^{s_0,p}(\Omega),$$

$$\|Tf\|_{k,p,\Omega} \lesssim N^{k-s_1} \|f\|_{s_1,p,\Omega}, \quad \forall f \in W^{s_1,p}(\Omega).$$

Now applying the theory of interpolation spaces (cf. Chapter 14 in (Brenner & Scott, 2008)), we obtain that T is also a bounded operator from $W^{s,p}(\Omega)$ into $W^{k,p}(\Omega)$ and

$$\begin{aligned} \|f - P_N f\|_{k,p,\Omega} &= \|Tf\|_{k,p,\Omega} \\ &\lesssim N^{(1-\theta)(k-s_0)+\theta(k-s_1)} \|f\|_{s,p,\Omega} = N^{k-s} \|f\|_{s,p,\Omega}. \end{aligned}$$

Therefore (8) for any non-integer s follows.

We finally consider the estimate of $|a_{\mathbf{n}}|$ for each multi-index $\mathbf{n}$ with $\mathbf{n} \leq 2N$. since the term $c_{\mathbf{k}}(f^*)$ for each $0 \leq \mathbf{k} \leq 2N$ appears at most $(N+1)^d$ times in the right-hand side of (12), and since

$$|c_{\mathbf{k}}(f^*)| \leq \frac{2}{(2\pi)^d} \int_{\mathbb{T}^d} |f^*(\mathbf{z})| d\mathbf{z} \leq \frac{2|\mathbb{T}^d|^{1-1/p}}{(2\pi)^d} \|f^*\|_{0,p,\mathbb{T}^d} \lesssim \|f\|_{0,p,\Omega},$$

we obtain $|a_{\mathbf{n}}| \lesssim \|f\|_{0,p,\Omega}$. This completes the proof of Lemma 2. $\square$

4.2. Proof of Theorem 1

Let $f \in W^{s,p}(\Omega)$ with $s > 0$ and $1 \leq p \leq \infty$, $k \in \mathbb{N}_0$ with $k \leq s$, and $N \in \mathbb{N}$. Then, by Lemma 2, there exists a polynomial $P_N f$ on Ω with degree at most $2N$ in each variable such that (8) holds.

As in (Leshno et al., 1993; Mhaskar, 1996), we will construct a shallow neural network approximation of each monomial term in the polynomial $P_N f$. We first note that, since σ is smooth and non-polynomial, there exists a real sequence $\{r_n\}_n$ such that $\sigma^{(n)}(r_n) \neq 0$ for any $n \in \mathbb{N}_0$. Let

$$\Psi(r; \mathbf{w}, \mathbf{x}) := \sigma(\mathbf{w} \cdot \mathbf{x} + r), \quad (r, \mathbf{w}, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^d \times \Omega.$$

Given multi-index $\mathbf{n} = (n_1, \dots, n_d)$ with $\mathbf{n} \leq 2N$, let $\Phi_{\mathbf{n}} : \mathbb{R}^d \times \Omega \rightarrow \mathbb{R}$ be given by

$$\Phi_{\mathbf{n}}(\mathbf{w}, \mathbf{x}) := \partial_{\mathbf{w}}^{\mathbf{n}} \Psi(r_{|\mathbf{n}|}; \mathbf{w}, \mathbf{x}), \quad (\mathbf{w}, \mathbf{x}) \in \mathbb{R}^d \times \Omega.$$

Then we have

$$\mathbf{x}^{\mathbf{n}} = (\sigma^{(|\mathbf{n}|)}(r_{|\mathbf{n}|}))^{-1} \Phi_{\mathbf{n}}(\mathbf{0}, \mathbf{x}), \quad \forall \mathbf{x} \in \Omega. \quad (13)$$

Next, consider the central finite difference approximation of $\Phi_{\mathbf{n}}(\cdot, \mathbf{x})$ at $\mathbf{w} = \mathbf{0}$

$$\Phi_{\mathbf{n},h}(\mathbf{x}) := (-h)^{-|\mathbf{n}|} \sum_{0 \leq \mathbf{m} \leq \mathbf{n}} (-1)^{|\mathbf{m}|} \binom{\mathbf{n}}{\mathbf{m}} \Psi(r_{|\mathbf{n}|}; h(\mathbf{m} - \mathbf{n}/2), \mathbf{x}),$$

where $h > 0$. Note that, for any multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$,

$$\partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n}}(\mathbf{w}, \mathbf{x}) = \partial_{\mathbf{w}}^{\mathbf{n}} (\partial_{\mathbf{x}}^{\mathbf{k}} \Psi)(r_{|\mathbf{n}|}; \mathbf{w}, \mathbf{x}),$$

$$\partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n},h}(\mathbf{x}) = (-h)^{-|\mathbf{n}|} \sum_{0 \leq \mathbf{m} \leq \mathbf{n}} (-1)^{|\mathbf{m}|} \binom{\mathbf{n}}{\mathbf{m}} (\partial_{\mathbf{x}}^{\mathbf{k}} \Psi)(r_{|\mathbf{n}|}; h(\mathbf{m} - \mathbf{n}/2), \mathbf{x}).$$

This implies that $\partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n},h}(\mathbf{x})$ is the central finite difference approximation of $\partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n}}(\cdot, \mathbf{x})$ at $\mathbf{w} = \mathbf{0}$. Thus, for any $\mathbf{x} \in \Omega$ and any multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$,

$$|\partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n}}(\mathbf{0}, \mathbf{x}) - \partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n},h}(\mathbf{x})| \leq \frac{h^2}{3n!} \sum_{|\mathbf{m}|=2} \sup_{\mathbf{w} \in [-2Nh, 2Nh]^d} |\partial_{\mathbf{w}}^{\mathbf{m}} \partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n}}(\mathbf{w}, \mathbf{x})|. \quad (14)$$

Note that

$$\begin{aligned} \partial_{\mathbf{w}}^{\mathbf{m}} \partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n}}(\mathbf{w}, \mathbf{x}) &= \partial_{\mathbf{x}}^{\mathbf{k}} \partial_{\mathbf{w}}^{\mathbf{n}+\mathbf{m}} \sigma(\mathbf{w} \cdot \mathbf{x} + r_{|\mathbf{n}|}) \\ &= \partial_{\mathbf{x}}^{\mathbf{k}} (\mathbf{x}^{\mathbf{n}+\mathbf{m}} \sigma^{(\mathbf{n}+\mathbf{m})}(\mathbf{w} \cdot \mathbf{x} + r_{|\mathbf{n}|})) \\ &= \sum_{\mathbf{i} \leq \mathbf{k}} (\partial_{\mathbf{x}}^{\mathbf{i}} \mathbf{x}^{\mathbf{n}+\mathbf{m}}) (\partial_{\mathbf{x}}^{\mathbf{k}-\mathbf{i}} \sigma^{(\mathbf{n}+\mathbf{m})}(\mathbf{w} \cdot \mathbf{x} + r_{|\mathbf{n}|})) \\ &= \sum_{\mathbf{i} \leq \mathbf{k}} \frac{(\mathbf{n}+\mathbf{m})!}{(\mathbf{n}+\mathbf{m}-\mathbf{i})!} \mathbf{x}^{\mathbf{n}+\mathbf{m}-\mathbf{i}} \mathbf{w}^{\mathbf{k}-\mathbf{i}} \sigma^{(\mathbf{n}+\mathbf{m}+\mathbf{k}-\mathbf{i})}(\mathbf{w} \cdot \mathbf{x} + r_{|\mathbf{n}|}). \end{aligned}$$

Let $C_{\mathbf{n},\sigma} = \|\sigma\|_{|\mathbf{n}|+2, \infty, [r_{|\mathbf{n}|}-1, r_{|\mathbf{n}|}+1]}$. Then, for any $h \in (0, 1/(2N))$, we have

$$\sup_{\mathbf{w} \in [-2Nh, 2Nh]^d} |\partial_{\mathbf{w}}^{\mathbf{m}} \partial_{\mathbf{x}}^{\mathbf{k}} \Phi_{\mathbf{n}}(\mathbf{w}, \mathbf{x})| \leq C_{\mathbf{n},\sigma} ([s] + 1)^d (2N)^{[s]d}. \quad (15)$$

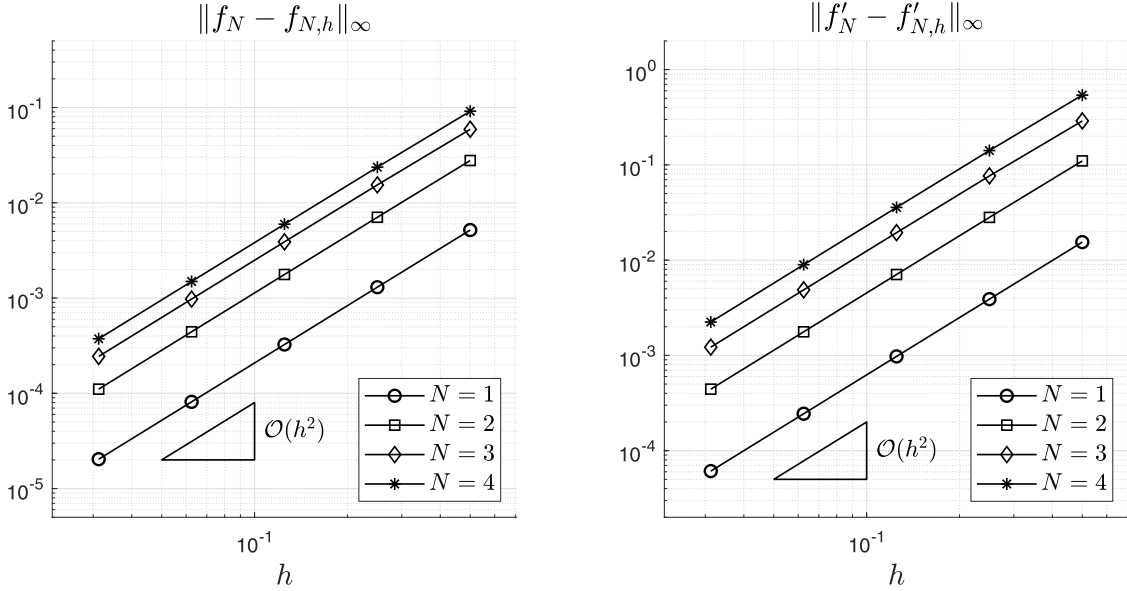


Fig. 1. The errors $\|f_N - f_{N,h}\|_\infty$ (left) and $\|f'_N - f'_{N,h}\|_\infty$ (right) versus h .

Now for each multi-index $\mathbf{n}$ with $|\mathbf{n}| \leq 2N$, let

$$h_{\mathbf{n}} = \min \left\{ \frac{1}{2N}, \left(\frac{6\mathbf{n}! \sigma^{(|\mathbf{n}|)}(r_{|\mathbf{n}|)})}{|\Omega| C_{\mathbf{n},\sigma} d(d+1)(\lceil s \rceil + 1)^d N^s (2N+1)^{(\lceil s \rceil + 1)d}} \right)^{1/2} \right\}.$$

Then, combining (13) (15) and (14), we have

$$\|\mathbf{x}^{\mathbf{n}} - (\sigma^{(|\mathbf{n}|)}(r_{|\mathbf{n}|)})^{-1} \Phi_{\mathbf{n},h_{\mathbf{n}}}\|_{k,p,\Omega} \leq (2N+1)^{-d} N^{-s}, \quad \forall 0 \leq k \leq s. \quad (16)$$

Note that $P_N f = \sum_{0 \leq |\mathbf{n}| \leq 2N} a_{\mathbf{n}} \mathbf{x}^{\mathbf{n}}$ with $|a_{\mathbf{n}}| \lesssim \|f\|_{0,p,\Omega}$. Now let

$$f_N(\mathbf{x}) := \sum_{0 \leq |\mathbf{n}| \leq 2N} a_{\mathbf{n}} (\sigma^{(|\mathbf{n}|)}(r_{|\mathbf{n}|)})^{-1} \Phi_{\mathbf{n},h_{\mathbf{n}}}(\mathbf{x}).$$

Then f_N is a neural network function having one hidden layer such that

$$\begin{aligned} \|P_N f - f_N\|_{k,p,\Omega} &\leq \sum_{0 \leq |\mathbf{n}| \leq 2N} |a_{\mathbf{n}}| \|\mathbf{x}^{\mathbf{n}} - (\sigma^{(|\mathbf{n}|)}(r_{|\mathbf{n}|)})^{-1} \Phi_{\mathbf{n},h_{\mathbf{n}}}\|_{k,p,\Omega} \lesssim N^{k-s} \|f\|_{0,p,\Omega}. \end{aligned}$$

Thus we have

$$\|f - f_N\|_{k,p,\Omega} \leq \|f - P_N f\|_{k,p,\Omega} + \|P_N f - f_N\|_{k,p,\Omega} \lesssim N^{k-s} \|f\|_{s,p,\Omega}.$$

Moreover, since

$$\{\mathbf{m} - \mathbf{n}/2 : \mathbf{m} \leq \mathbf{n}, \mathbf{n} \leq 2N\} \subseteq \{\mathbf{k}/2 : \mathbf{k} \in \mathbb{Z}^d \cap [-2N, 2N]^d\},$$

the hidden layer in f_N has width at most $(4N+1)^d$. This proves (1).

We next prove (2). According to the density of $C^\infty(\Omega)$ in $W^{s,p}(\Omega)$ (cf. Grisvard (2011), Wloka (1987)), given $\epsilon > 0$, there exists $\psi_\epsilon \in C^\infty(\Omega)$ such that

$$\|f - \psi_\epsilon\|_{s,p,\Omega} \leq \frac{1}{2}\epsilon.$$

Let $\lceil s \rceil$ denote the smallest integer not smaller than s . Since ψ_ϵ is smooth, it follows from (1), where f is replaced with ψ_ϵ and s is replaced with $\lceil s \rceil + 1$, that there exists a shallow neural network function f_ϵ with activation σ such that

$$\|\psi_\epsilon - f_\epsilon\|_{s,p,\Omega} \leq \|\psi_\epsilon - f_\epsilon\|_{\lceil s \rceil,p,\Omega} \leq \frac{1}{2}\epsilon.$$

Now (2) follows by combining the last two inequalities, and we complete the proof of Theorem 1.

Remark 4. Based on the construction in the proof, the biases in the first layer of f_N are bounded above by $\max_{0 \leq |\mathbf{n}| \leq 2N} |r_{|\mathbf{n}|}|$, while the bias in the

second layer of f_N is fixed at a_0 . The weights in the first layer of f_N are bounded above by hN , where

$$h = \max_{|\mathbf{n}| \leq 2N} h_{\mathbf{n}} \approx ((2N)!)^{1/2} N^{1-s/2} (2N+1)^{-(\lceil s \rceil + 1)d/2}.$$

while the weights in the second layer of f_N are bounded above by

$$\max_{|\mathbf{n}| \leq 2N} \frac{((2N)!)^d |a_{\mathbf{n}}|}{h^{2Nd} |\sigma^{(\mathbf{n})}(r_{|\mathbf{n}|})|} \lesssim \|f\|_{0,p,\Omega} C^N ((2N)!)^{d(1-N)} N^{(-2+s+2(\lceil s \rceil + 1)d)dN},$$

where C is a positive constant depending only on s, p, d , and σ . On the other hand, the constant $C(s, p, d)$ in (1) is bounded above by $C_1 + C_2$, where C_1 is the hidden constant in (8), and C_2 is the constant from the inequality $\max_{|\mathbf{n}| \leq 2N} |a_{\mathbf{n}}| \leq C_2 \|f\|_{0,p,\Omega}$, both of which depend only on s, p , and d .

4.3. Proof of Theorem 2

Consider the following statement for $L \in \mathbb{N}$:

⊗ For each $N \in \mathbb{N}$, there exists a neural network f_N with smooth non-polynomial activation functions contained in $W^{[s]+2,\infty}(\mathbb{R})$ and with L hidden layers, one having width at most $(4N+1)^d$ and the other having width at most 2, such that

$$\|f - f_N\|_{k,p,\Omega} \leq c_0 c_L N^{k-s} \|f\|_{s,p,\Omega}, \quad \forall k \in \mathbb{N}_0 \text{ with } k \leq s,$$

where $c_0 \geq 1$ is a positive constant depending only on s, p, d and σ , and c_L is given by $c_L = \prod_{\ell=1}^L (1 + \ell^{-2})$.

Note that, for any $L \in \mathbb{N}$,

$$c_L = \prod_{\ell=1}^L \left(1 + \frac{1}{\ell^2} \right) \leq \exp \left(\sum_{\ell=1}^L \frac{1}{\ell^2} \right) < \infty.$$

Thus (3) holds if ⊗ holds for any $L \in \mathbb{N}$. Furthermore, (4) can be shown proceeding as in the proof of (2). Therefore it is enough to prove that ⊗ holds for any $L \in \mathbb{N}$.

To do this, we will use the mathematical induction on L . Since the case when $L = 1$ has been already proved in Theorem 1, it suffices to prove that if ⊗ with $L = \ell$ holds for some $\ell \geq 1$ then ⊗ with $L = \ell + 1$ also holds. Suppose that there exists a neural network function $f_{N,\ell}$ with smooth non-polynomial activation functions contained in $W^{[s]+2,\infty}(\mathbb{R})$ and with ℓ hidden layers, one having width at most $(4N+1)^d$ and the other having width at most 2, such that

$$\|f - f_{N,\ell}\|_{k,p,\Omega} \leq c_0 c_\ell N^{k-s} \|f\|_{s,p,\Omega}, \quad \forall k \in \mathbb{N}_0, k \leq s. \quad (17)$$

In order to construct a neural network approximation of f having $\ell + 1$ hidden layers, we generalize the idea used in [Theorem 1](#). Recall that, since σ is smooth and non-polynomial, there exists a real sequence $\{r_n\}_n$ such that $\sigma^{(n)}(r_n) \neq 0$ for any $n \in \mathbb{N}_0$. Let Ψ and Φ be the real-valued smooth functions on $\mathbb{R} \times \Omega$ given by

$$\Psi(w, \mathbf{x}) = \sigma(w f_{N, \ell}(\mathbf{x}) + r_1), \quad \Phi(w, \mathbf{x}) = \partial_w \Psi(w, \mathbf{x}), \quad (w, \mathbf{x}) \in \mathbb{R} \times \Omega.$$

Then we have

$$f_{N, \ell}(\mathbf{x}) = (\sigma'(r_1))^{-1} \Phi(0, \mathbf{x}), \quad \forall \mathbf{x} \in \Omega. \quad (18)$$

Consider the central finite difference approximation of $\Phi(\cdot; \mathbf{x})$ at $w = 0$

$$\Phi_h(\mathbf{x}) := (2h)^{-1} (\Psi(h, \mathbf{x}) - \Psi(-h, \mathbf{x})),$$

where $h > 0$ will be determined later. Note that, for any multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$,

$$\begin{aligned} \partial_{\mathbf{x}}^{\mathbf{k}} \Phi(w, \mathbf{x}) &= \partial_w (\partial_{\mathbf{x}}^{\mathbf{k}} \Psi)(w, \mathbf{x}), \\ \partial_{\mathbf{x}}^{\mathbf{k}} \Phi_h(\mathbf{x}) &= (2h)^{-1} ((\partial_{\mathbf{x}}^{\mathbf{k}} \Psi)(h, \mathbf{x}) - (\partial_{\mathbf{x}}^{\mathbf{k}} \Psi)(-h, \mathbf{x})). \end{aligned}$$

This implies that $\partial_{\mathbf{x}}^{\mathbf{k}} \Phi_h(\mathbf{x})$ is the central finite difference approximation of $\partial_{\mathbf{x}}^{\mathbf{k}} \Phi(\cdot, \mathbf{x})$ at $w = 0$, and

$$|\partial_{\mathbf{x}}^{\mathbf{k}} \Phi(0, \mathbf{x}) - \partial_{\mathbf{x}}^{\mathbf{k}} \Phi_h(\mathbf{x})| \leq \frac{h^2}{6} \sup_{w \in [-h, h]} |(\partial_w^3 \partial_{\mathbf{x}}^{\mathbf{k}} \Psi)(w, \mathbf{x})|, \quad (19)$$

for any $\mathbf{x} \in \Omega$ and multi-index $\mathbf{k}$ with $|\mathbf{k}| \leq s$. Here, note that Ψ is smooth on $\mathbb{R} \times \Omega$, and Ω is bounded. Let

$$h_{\ell+1} = \min \left\{ 1, \left(\frac{|\sigma'(r_1)| c_0 c_{\ell} \|f\|_{s, p, \Omega}}{d^s (\ell + 1)^2 |\Omega| \|\Psi\|_{W^{[s]+3, \infty}([-1, 1] \times \Omega)}} N^{-s} \right)^{1/2} \right\}.$$

We then obtain from (18) and (19) that

$$\begin{aligned} \|f_{N, \ell} - (\sigma'(r_1))^{-1} \Phi_{h_{\ell+1}}\|_{k, p, \Omega} \\ = (\sigma'(r_1))^{-1} \|\Phi(0, \cdot) - \Phi_h\|_{k, p, \Omega} \leq \frac{c_0 c_{\ell}}{(\ell + 1)^2} N^{-s} \|f\|_{k, p, \Omega}. \end{aligned}$$

Now let us define

$$f_{N, \ell+1}(\mathbf{x}) := (\sigma'(r_1))^{-1} \Phi_{h_{\ell+1}}(\mathbf{x}),$$

which is indeed a neural network function having $\ell + 1$ hidden layers, one having width at most $(4N + 1)^d$ and the other having width at most 2, such that

$$\begin{aligned} \|f - f_{N, \ell+1}\|_{k, p, \Omega} &\leq \|f - f_{N, \ell}\|_{k, p, \Omega} + \|f_{N, \ell} - f_{N, \ell+1}\|_{k, p, \Omega} \\ &\leq c_0 c_{\ell} N^{k-s} \|f\|_{s, p, \Omega} + (\sigma'(r_1))^{-1} \|\Phi(0, \cdot) - \Phi_{h_{\ell+1}}\|_{k, p, \Omega} \\ &\leq c_0 c_{\ell} \left(1 + \frac{1}{(\ell + 1)^2} \right) N^{k-s} \|f\|_{s, p, \Omega} = c_0 c_{\ell+1} N^{k-s} \|f\|_{s, p, \Omega}. \end{aligned}$$

This proves $\textcircled{\ast}$ with $L = \ell + 1$, and we complete the proof of [Theorem 2](#).

Remark 5. According to the proof, the weights and biases in the first layer of $f_{N, \ell+1}$ coincide with those of the shallow neural network constructed in the proof of [Theorem 1](#). However, it is challenging to explicitly estimate the scale of the weights and biases in the subsequent layers, as they depend on $\|\Psi\|_{W^{[s]+3, \infty}([-1, 1] \times \Omega)}$. On the other hand, the constant $C(s, p, d)$ in (3) is uniformly bounded above by

$$C(s, p, d) \leq C_0 \exp \left(\sum_{\ell=1}^{\infty} \frac{1}{\ell^2} \right) \leq 6C_0,$$

where C_0 is the constant $C(s, p, d)$ in (1).

4.4. Proof of [Theorem 3](#)

We first find the polynomial $Q_N f$ of degree at most $N + N_0$ (with positive integer N_0 depending only on s, d and R_f) such that

$$\|f - Q_N f\|_{s, \infty, \Omega} \leq C'(s, d, R_f) \exp(-N), \quad (20)$$

where $C'(s, d, R_f) > 0$ is a constant depending only on s, d and R_f . According to Taylor's theorem and $\Omega = [-1, 1]^d$, we have

$$\|f - Q_N f\|_{s, \infty, \Omega} \leq \sum_{|\mathbf{n}|=N+N_0+1-s} \frac{|f|_{N+N_0+1, \infty, \Omega}}{\mathbf{n}!},$$

where $Q_N f$ is the polynomial of degree at most $N + N_0$ defined on Ω such that

$$Q_N f(\mathbf{x}) = \sum_{|\mathbf{n}| \leq N+N_0} \frac{\partial^{\mathbf{n}} f(\mathbf{0})}{\mathbf{n}!} \mathbf{x}^{\mathbf{n}}, \quad \mathbf{x} \in \Omega.$$

By the multinomial theorem and the assumption $|f|_{N+N_0+1, \infty, \Omega} \leq R_f^{N+N_0+1}$,

$$\sum_{|\mathbf{n}|=N+N_0+1-s} \frac{|f|_{N+N_0+1, \infty, \Omega}}{\mathbf{n}!} \leq \frac{(d R_f)^{N+N_0+1}}{(N + N_0 - s + 1)!}.$$

If $N_0 = s + 3 \lceil d R_f \rceil$, then

$$\frac{(d R_f)^{N+N_0+1}}{(N + N_0 - s + 1)!} \leq \frac{(d R_f)^{N_0}}{(N_0 - s)!} \exp(-N).$$

Therefore we obtain (20) with $C'(s, d, R_f) = (d R_f)^{N_0} / (N_0 - s)!$.

The next step is to construct the neural network function f_N with L hidden layers, one having width at most $(2N + 2s + 6 \lceil d R_f \rceil + 1)^d$ and the other having width at most 2 such that

$$\|Q_N f - f_N\|_{s, \infty, \Omega} \leq C(s, d, R_f) \exp(-N),$$

where $C(s, d, R_f) > 0$ is a constant depending only on s, d and R_f . This can be obtained by proceeding as in the proof of [Theorems 1](#) and [2](#).

4.5. Numerical example

Here we provide a numerical experiment to confirm the simultaneous approximation of the monomials by shallow neural networks, which is the main idea of the proof of the main results.

Let $d = 1$ and consider the monomial $f_N(x) = x^N$ for $N = 1, \dots, 4$. Following the construction given in [Section 4.3](#), we can obtain its shallow neural network approximation as follows:

$$f_{N, h} = (\sigma^{(N)}(r_N))^{-1} (-h)^{-N} \sum_{m=0}^N (-1)^m \binom{N}{m} \sigma(h(m - N/2)x + r_N),$$

where $h > 0$ and r_N is chosen such that $\sigma^{(N)}(r_N) \neq 0$. We use the sigmoid activation function $\sigma(x) = (1 + \exp(-x))^{-1}$ and select the bias shifts as $(r_1, r_2, r_3, r_4) = (0, 1, 0, 1)$, which ensures that $\sigma^{(N)}(r_N) \neq 0$ for each $N = 1, \dots, 4$ (as $\sigma^{(2)}(0) = \sigma^{(4)}(0) = 0$, non-zero shifts are required for $N = 2, 4$).

Under the setting above, we then compute the approximation errors $\|f_N - f_{N, h}\|_{\infty}$ and $\|f'_N - f'_{N, h}\|_{\infty}$ for each $h = 1/2^1, \dots, 1/2^5$, where the maximum norm $\|\cdot\|_{\infty}$ is estimated over a uniform grid on $[-1, 1]$:

$$\|v\|_{\infty} := \max \{ |v(x)| : x = i\delta, i \in \mathbb{Z} \cap [-100, 100] \}, \quad \delta = 0.01.$$

The results are reported in [Fig. 1](#). As shown, the errors decay with rate $\mathcal{O}(h^2)$, which is consistent with the theoretical expectation from (14).

5. Conclusion

In this paper, we have studied the approximation capabilities of neural networks with smooth non-polynomial activation and with a fixed depth, including the shallow networks, for the Sobolev-regular functions with respect to the Sobolev norm. Namely, we have proved that such neural networks, one layer having width $\mathcal{O}(N^d)$ and the others having width $\mathcal{O}(1)$ can achieve the $W^{k, p}$ -norm error bound $\mathcal{O}(N^{k-s})$ for the $W^{s, p}$ -regular functions.

As future work, we wish to reduce the scale of the weights and biases while maintaining the error bounds and investigate whether our results can be improved so that the weights and biases are encodable as in ([Gühring & Raslan, 2021](#)). Another possible direction is to improve our results so that the error bound decreases as the depth increases.

It is also worth noting that our results were established on bounded domains. However, extension of our results to unbounded domains (e.g., $\mathbb{R}^d$) seems challenging, since our proofs fundamentally rely on polynomial approximation on bounded domains and the periodic extension of the functions in the Sobolev spaces on $[-1, 1]^d$, which do not hold globally on unbounded domains. To address this, we may need to introduce weighted Sobolev spaces or adopt approximation results by the functions different from the polynomials. We leave it for a future investigation.

CRediT authorship contribution statement

Hyeokjoo Park: Writing – review & editing, Writing – original draft, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by [National Research Foundation of Korea](#) (NRF) grant funded by the Korea Government (MSIT) (No. [NRF-RS-2025-00513210](#)).

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